Class 11 Physics SVPHS – Assignment

Laws of Motion, Work

Q1. A force of $40\,\mathrm{N}$ acts on a $5\,\mathrm{kg}$ block resting on a frictionless horizontal surface. Find its acceleration.

Q2. A $12\,\mathrm{kg}$ object accelerates at $3\,\mathrm{m/s^2}$. Calculate the net force acting on it.

Q3. A $6\,\mathrm{kg}$ block experiences two horizontal forces of $25\,\mathrm{N}$ and $10\,\mathrm{N}$ acting in opposite directions. Find the acceleration.

Q4. A $4\,\mathrm{kg}$ body moving at $8\,\mathrm{m/s}$ comes to rest in $4\,\mathrm{s}$ due to a constant retarding force. Calculate the retarding force.

Q5. A $10\,\mathrm{kg}$ block is pulled horizontally by a force of $60\,\mathrm{N}$. If the acceleration is $4\,\mathrm{m/s^2}$, calculate the frictional force.

Q6. A $15\,\mathrm{kg}$ body is pushed with a force of $90\,\mathrm{N}$ and accelerates at $5\,\mathrm{m/s^2}$. Determine the opposing force.

Q7. A $20\,\mathrm{kg}$ cart accelerates at $2.5\,\mathrm{m/s^2}$ under a horizontal force. Calculate the applied force.

Q8. A body of mass $8\,\mathrm{kg}$ moves with uniform velocity despite a horizontal force of $24\,\mathrm{N}$ acting on it. Calculate the frictional force.

Q9. A $10\,\mathrm{kg}$ block hangs freely from a light string. Find the tension in the string.

Q10. Two masses of $6\,\mathrm{kg}$ and $4\,\mathrm{kg}$ are connected by a light string over a smooth pulley. Find the acceleration of the system.

Q11. For the system in Question 10, calculate the tension in the string.

Q12. A $12\,\mathrm{kg}$ block hangs from a rope in an elevator accelerating upward at $2\,\mathrm{m/s^2}$. Find the tension.

Q13. A $15\,\mathrm{kg}$ object is suspended in an elevator moving downward with acceleration $3\,\mathrm{m/s^2}$. Find the apparent weight.

Q14. Calculate the weight of a $75\,\mathrm{kg}$ person on Earth. Take $g = 9.8\,\mathrm{m/s^2}$.

Q15. A $20\,\mathrm{kg}$ block rests on a rough horizontal surface with coefficient of friction $\mu = 0.3$. Find the limiting friction.

Q16. A $15\,\mathrm{kg}$ block is pulled with a horizontal force of $70\,\mathrm{N}$. The coefficient of kinetic friction is 0.2. Find the acceleration.

Q17. A $10\,\mathrm{kg}$ block rests on a rough inclined plane of 30°. The coefficient of friction is 0.4. Determine whether the block will slide.

Q18. Calculate the frictional force acting on a $25\,\mathrm{kg}$ block if the coefficient of friction is 0.25.

Q19. A 50 kg crate is pushed with 180 N on a rough floor. If $\mu = 0.3$, determine whether it moves.

Q20. A 30 kg block slides with constant velocity on a rough surface under a pulling force of 90 N. Find the coefficient of kinetic friction.

Q21. A car travels at 20 m/s on a level circular road of radius 80 m. Find the required centripetal force if the car's mass is 1000 kg.

Q22. A road is banked at an angle of $15°$ with radius 60 m. Calculate the safe speed without friction.

Q23. A motorcycle of mass 200 kg moves around a circular path of radius 40 m at 18 m/s. Calculate the centripetal force.

Q24. Find the angle of banking for a road of radius 100 m designed for a speed of 54 km/h.

Q25. A 2 kg object moving at 6 m/s collides and sticks to a 3 kg object at rest. Find their common velocity.

Q26. A 1000 kg car moving at 20 m/s collides with a 1500 kg stationary car and they move together. Find their common velocity.

Q27. A 5 kg rifle fires a 0.02 kg bullet at 500 m/s. Calculate the recoil velocity of the rifle.

Q28. A 0.15 kg ball moving at 20 m/s strikes a wall and rebounds at 15 m/s. Find the change in momentum.

Q29. A car starts from rest and accelerates uniformly at $2\,\text{m/s}^2$ for 8 s. Find the final velocity.

Q30. A train moving at 12 m/s accelerates uniformly at $1.5\,\text{m/s}^2$ for 10 s. Calculate the distance travelled.

Q31. A body moving at 25 m/s comes to rest with uniform retardation of $5\,\text{m/s}^2$. Find the stopping distance.

Q32. A particle starts with velocity 10 m/s and accelerates uniformly at $3\,\text{m/s}^2$. Find its velocity after travelling 20 m.

Q33. A force of 50 N acts on a body causing a displacement of 8 m in the same direction. Calculate the work done.

Q34. A 15 kg object is lifted vertically through a height of 6 m. Calculate the work done against gravity.

Q35. A machine performs 24 000 J of work in 40 s. Calculate its power.

Q36. A 200 W motor operates for 3 minutes. Calculate the work done.

Q37. Find the kinetic energy of a 5 kg body moving at 12 m/s.

Q38. Calculate the gravitational potential energy of a 10 kg object placed 15 m above the ground.

Q39. A 4 kg object falls freely from a height of 20 m. Calculate its kinetic energy just before reaching the ground.

Q40. A 1500 kg car moving at 20 m/s is brought to rest by brakes. Calculate the work done by the braking force.

Useful Constants

- Gravitational acceleration:
$$g = 9.8\,\mathrm{m/s^2}$$
- For numerical calculations:
$$\pi = 3.14$$

Enjoy your Vacation and **Email the assignment to:**

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